

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2018

EC248/EC210.01 Analog Communication

Date: 09.05.2018, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) An audio frequency signal is $20\sin(2\pi 1000t)$ used to amplitude modulate a carrier wave of $70\sin(2\pi 10^6 t)$. Calculate [07]

- i. Modulation Index
- ii. Sideband frequencies
- iii. Amplitude of each sideband frequencies
- iv. Bandwidth Required
- v. Total power delivered to load of 800Ω
- vi. Transmission efficiency

Q - 2 (a) Derive the mathematical equation of effective modulation index, voltage and current for several sinusoidal Amplitude Modulated (AM) wave. [07]

(b) What is the relation between total power, unmodulated carrier power and modulation index? Derive its mathematical expressions. [07]

OR

Q - 2 (a) How broadcast Amplitude Modulated (AM) transmitter transmit information signal in practical radio receiver? Describe its step process. [07]

(b) Draw and explain the block diagram of low level and high level amplitude modulator (AM) transmitter. [07]

Q - 3 (a) Draw and explain the circuit diagram of ring or lattice modulator for DSBSC (double sideband suppressed carrier) generation. [07]

(b) Draw and explain the block diagram of “phase shift” method for single sideband suppressed carrier (SSBSC) generation. [07]

OR

Q - 3 (a) Draw and explain the circuit diagram of balance modulator using FET (field effect transistor) for DSBSC (double sideband suppressed carrier) generation. [07]

- (b) What is the need for modulation? [03]
What are the advantages of DSB-SC and SSB-SC? [04]

SECTION – II

- Q - 4 (a) Draw and explain FET reactance modulator with necessary circuit diagram. [07]
Q - 5 (a) Draw and explain the block diagram of superheterodyne receiver with necessary detail. [07]
(b) Explain foster seeley discriminator with necessary block diagram and phasor diagram. [07]

OR

- Q - 5 (a) How receiver tracking system works? Describe various types of receiver tracking system. [07]
(b) Explain frequency modulation with waveform. Also derive mathematical equation of frequency modulated (FM) wave. [07]
Q - 6 (a) A receiver tunes signal from 550 KHz to 1600 KHz with an IF of 455 KHz. Find the frequency tuning ranges and capacitor tuning ranges for oscillator section and for RF section. [04]
(b) Explain image rejection in superheterodyne receiver. [03]
(c) Compare various technical aspects of frequency modulation (FM) system with amplitude modulation (AM) system. [07]

OR

- Q - 6 (a) What is double spotting? Also explain spurious responses in detail. [07]
(b) Determine appropriate multiplier values and oscillator frequency so as to provide an output FM signal having carrier frequency of 106MHz with frequency deviation of 60KHz. As same the input signal to be an FM signal having carrier frequency of 9 MHz and frequency deviation of 10KHz. [05]
(c) Define carson's rule for frequency modulation (FM). [02]
